

# AI-Powered Policy Compliance Intelligence Assistant

## Overview

Large organizations maintain extensive internal policy documents covering areas such as HR policies, cybersecurity guidelines, and data governance rules. These documents are essential for ensuring compliance, but they are often long and complex, making it difficult for employees to quickly locate relevant information during daily work.

Employees frequently ask questions such as:

- *“Can I continue working remotely while traveling to another country?”*
- *“If I accidentally see data that does not belong to my project, should I report it?”*
- *“Are there any restrictions when sharing customer data from different regions with another internal team?”*

In modern enterprise environments, employees must often interpret multiple overlapping frameworks, including internal policies, cybersecurity standards such as NIST, and regional data protection regulations like GDPR. Manually searching through these documents can be time-consuming and may lead to uncertainty or incorrect decisions.

## Objective

The objective of this project is to build an AI-powered policy compliance intelligence assistant that allows employees to ask natural-language questions and receive accurate answers grounded in official policy documents and regulatory guidelines.

The system should provide clear explanations, references to relevant policy sections, and context-aware compliance guidance, helping employees understand the correct course of action and potential compliance risks in different situations.

## Key Capabilities

### **Natural Language Policy Queries**

Employees can ask questions in everyday language instead of manually browsing large policy documents.

### **Policy Interpretation**

The system explains complex policy rules in a clear and understandable way.

### **Context-Aware Compliance Guidance**

Responses consider the employee's situation and provide guidance aligned with company policies and regulations.

### **Regulatory References**

The assistant highlights relevant sections of internal policies or external compliance frameworks.

### **Risk and Escalation Awareness**

The system identifies potential compliance risks and recommends whether issues should be reported or escalated.

### **Requirement 1 (Basic)**

- Basic RAG with policy document chunking
- Semantic search for policy queries
- Simple Q&A agent for compliance questions
- Metadata filtering (policy category, framework)
- Input guardrails for valid queries
- Citation of policy sources
- Basic compliance validation
- Hybrid retrieval using keyword search and vector embeddings

Expose the core functionality through API endpoint so that users or other applications can interact with the system.

### **Requirement 2 (Advanced)**

- DeepEval testing for policy retrieval accuracy
- Context precision metrics for compliance answers
- Reranking policies by relevance and authority
- LLM-as-judge for compliance interpretation quality
- Token usage tracking for policy analysis
- Performance benchmarking across policy frameworks

Build a simple front-end interface to demonstrate how an end user can interact with the service.

### **Advanced Compliance Intelligence**

- Cross-document policy reasoning across multiple regulatory frameworks
- Conflict detection when different policies provide contradictory guidance
- Structured compliance explanations with policy clause references

### **Multi-Agent Compliance System**

**Implement a multi-agent architecture for policy reasoning:**

- Policy Retrieval Agent – retrieves relevant policy sections
- Policy Interpretation Agent – interprets regulatory language
- Compliance Checker Agent – verifies if the requested action is compliant
- Risk Assessment Agent – evaluates the severity of potential violations
- Recommendation Agent – suggests compliant alternatives or escalation paths

### **Additional System Capabilities**

- Agent handoff for cross-framework compliance reasoning
- Agent-to-Agent communication between regulatory domain experts
- Dynamic compliance scoring for employee actions
- Compliance analytics dashboard showing common policy queries and violations

## **Deliverables**

### **1. Architecture Diagram**

Include a high-level view of your system in JPEG or PDF.

Show how data flows through:

- policy ingestion pipeline
- document chunking and embedding generation
- hybrid retrieval layer
- multi-agent reasoning system
- final response generation

### **2. Design**

Articulate system design decisions taken and the trade-offs made, such as:

- vector database selection
- chunking strategy for policy documents
- hybrid search vs semantic-only search
- agent orchestration architecture
- guardrail design for compliance accuracy

### **3. Full Executable Code (Microservice)**

Focus on clarity and modularity.

Provide a README explaining:

- Project setup (how to install or run)
- How policy documents are ingested into the system
- Sample usage (e.g., a test query and the resulting compliance guidance)
- Example of compliance risk explanation generated by the system

### **4. Panel Presentation (10 minutes)**

Demo of your working solution outlining the design and approach (8 minutes).

Example demo flow:

- User enters a compliance question
- System retrieves relevant policies
- Multi-agent reasoning evaluates compliance
- System generates policy-grounded answer with risk explanation

Q&A with the panel (2 minutes).

### **Dataset Name**

NIST Cybersecurity Framework Policy Documents

## Primary Link

<https://www.cisecurity.org/-/media/project/cisecurity/cisecurity/data/media/files/uploads/2024/08/cis-ms-isac-nist-cybersecurity-framework-policy-template-guide-2024.pdf>

## Alternative Links

<https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.29.pdf>  
<https://heimdalsecurity.com/blog/wp-content/uploads/NIST-Cybersecurity-Framework-Policy-Template-Guide-1.pdf>

## Format

PDF

## Key Fields

Policy text  
categories  
subcategories  
NIST functions

## Reason

Structured policy guidelines and compliance rules suitable for document chunking, semantic retrieval, and regulatory reasoning across multiple frameworks.